Civix: Digital Civic Engagement & Petition Platform

Project Statement:
Civix enables citizens to engage in local governance through petitions, voting, and tracking officials' responses. It fosters community-driven advocacy by allowing geo-targeted issues and public sentiment polling.

Outcomes:
Users can create and sign petitions.
Petitions are categorized and geo-tagged.
Users can vote in public sentiment polls.
Officials and admins track and respond to public interest.
Transparency and accountability reports are generated.

Modules:
Module A: User Authentication and Roles
Module B: Petition Management
Module C: Public Polling and Voting
Module D: Governance Dashboard and Reporting

Milestone 1: Weeks 1-2 - User Roles & Verification
Register as a citizen or official.
Geo-location and optional ID verification.
Secure login/authentication.
Output:
Authentication Screens.
Dashboard screen

Milestone 2: Weeks 3-4 - Petition Management
Users create, edit, and sign petitions.
Petitions are filterable by location and category.
Status: active, under review, closed.
Output:
Petition creation interface.
Petition listing and filter view.

Milestone 3: Weeks 5-6 - Polling and Feedback
Issue-specific polls.
Public voting interface.
Live sentiment graphs.

Output:
Poll creation tool.
Results dashboard.

Milestone 4: Weeks 7-8 - Governance Response and Reports
Officials see petitions in their locality.
Respond with comments or status updates.
Generate monthly civic engagement reports.
Output:
Reports dashboard.
Exportable reports.

Database Schema:
Users
- id: INTEGER (Primary Key, Auto-increment)
- name: VARCHAR(100)
- email: VARCHAR(150), UNIQUE
- password: VARCHAR(255)
- role: ENUM('citizen', 'official')
- location: VARCHAR(100)

Petitions
- id: INTEGER (Primary Key, Auto-increment)
- creator_id: INTEGER (Foreign Key → Users.id)
- title: VARCHAR(200)
- description: TEXT
- category: VARCHAR(100)
- location: VARCHAR(100)
- status: ENUM('active', 'under_review', 'closed')

Signatures
- id: INTEGER (Primary Key, Auto-increment)
- petition_id: INTEGER (Foreign Key → Petitions.id)
- user_id: INTEGER (Foreign Key → Users.id)
- timestamp: DATETIME

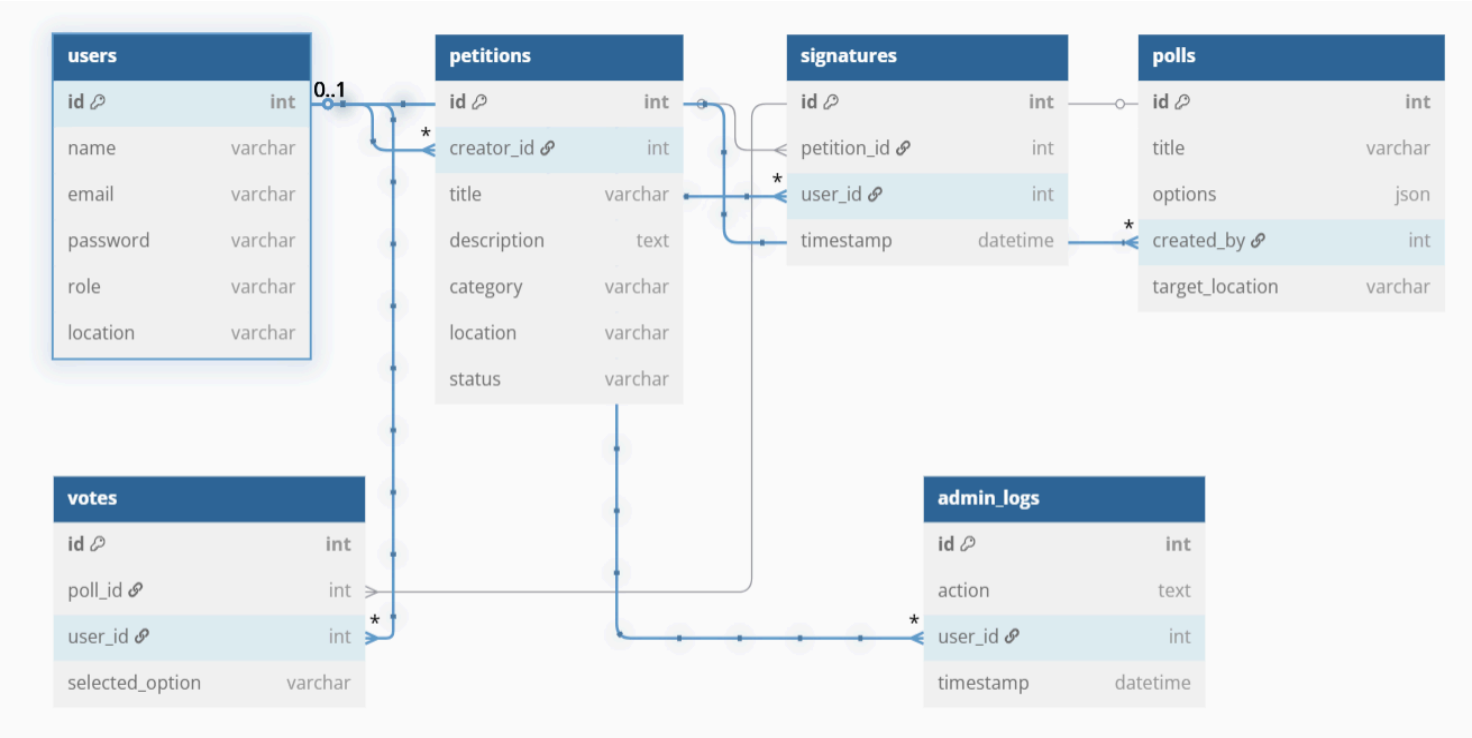
Polls
- id: INTEGER (Primary Key, Auto-increment)
- title: VARCHAR(200)
- options: JSON

- created_by: INTEGER (Foreign Key → Users.id)
- target_location: VARCHAR(100)

Votes
- id: INTEGER (Primary Key, Auto-increment)
- poll_id: INTEGER (Foreign Key → Polls.id)
- user_id: INTEGER (Foreign Key → Users.id)
- selected_option: VARCHAR(100)

AdminLogs
- id: INTEGER (Primary Key, Auto-increment)
- action: TEXT
- user_id: INTEGER (Foreign Key → Users.id)
- timestamp: DATETIME

Database Diagram:

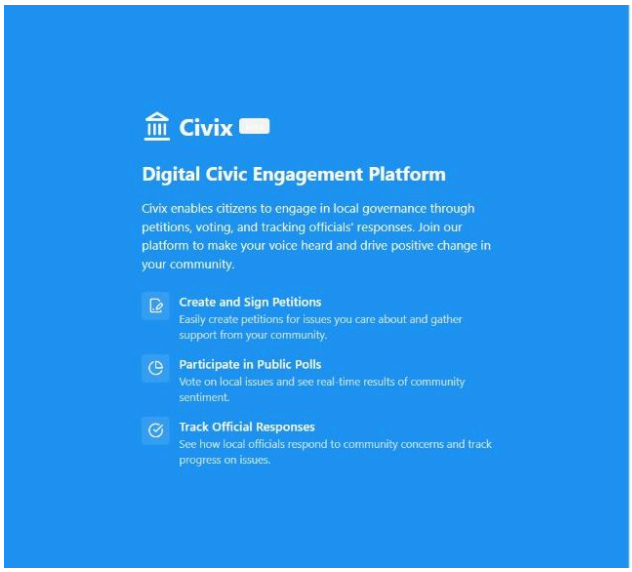


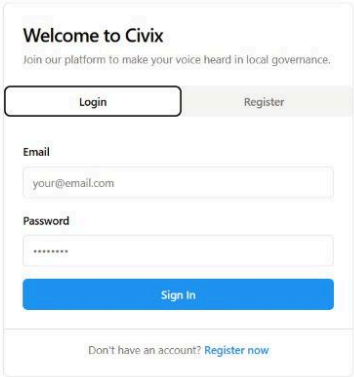
Evaluation Criteria:

Milestone 1: Verified role-based user system operational.
Milestone 2: Petition creation and signing implemented.
Milestone 3: Polling and sentiment tracking live.
Milestone 4: Government response tools and reporting functional.
Version control and tagging were completed for each milestone.

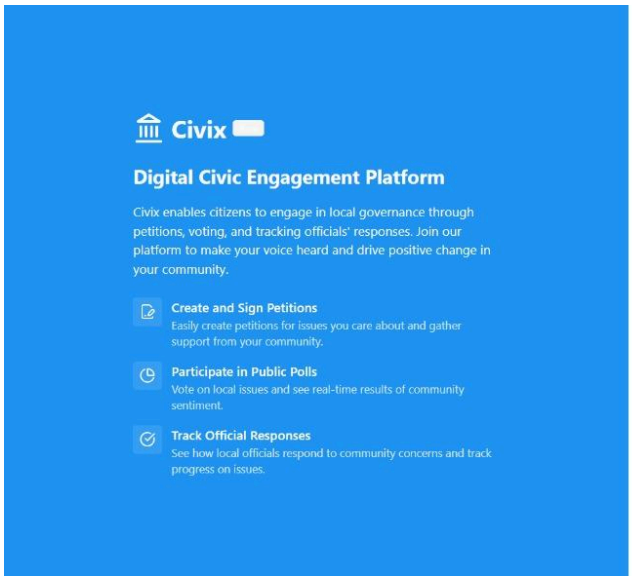
Mockups:

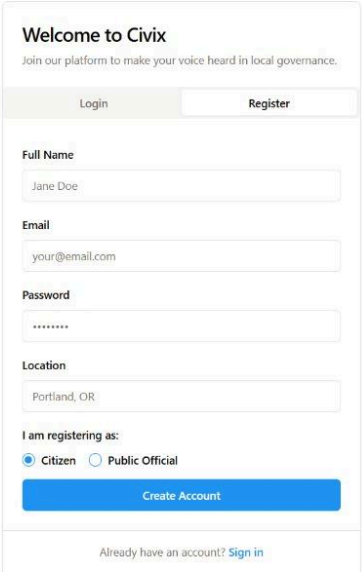
Login screen -



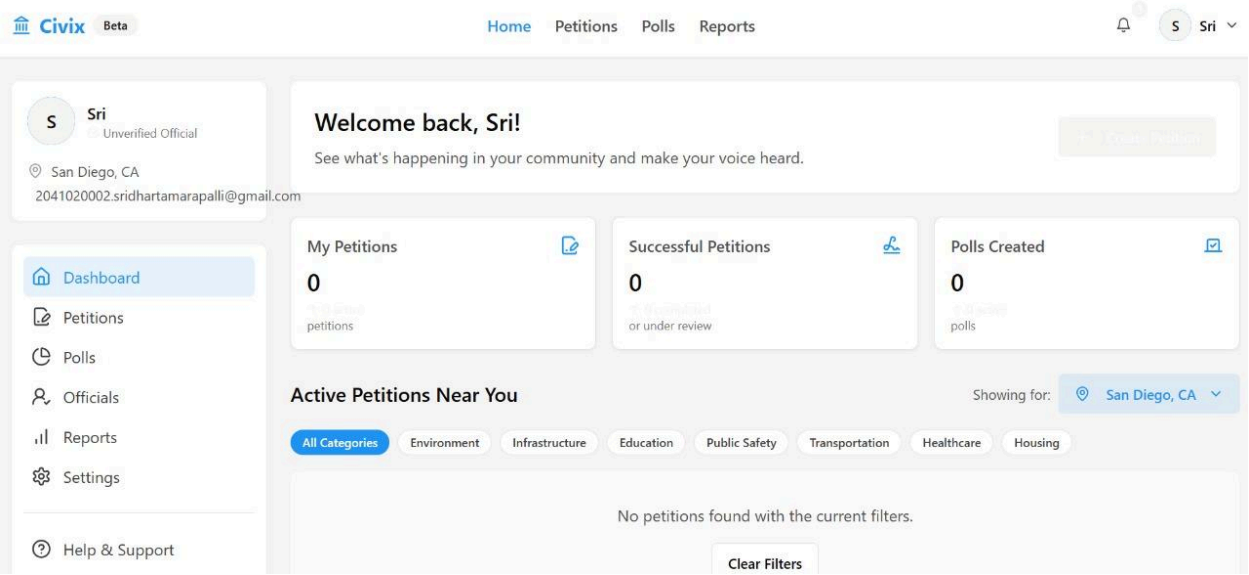


Register screen -

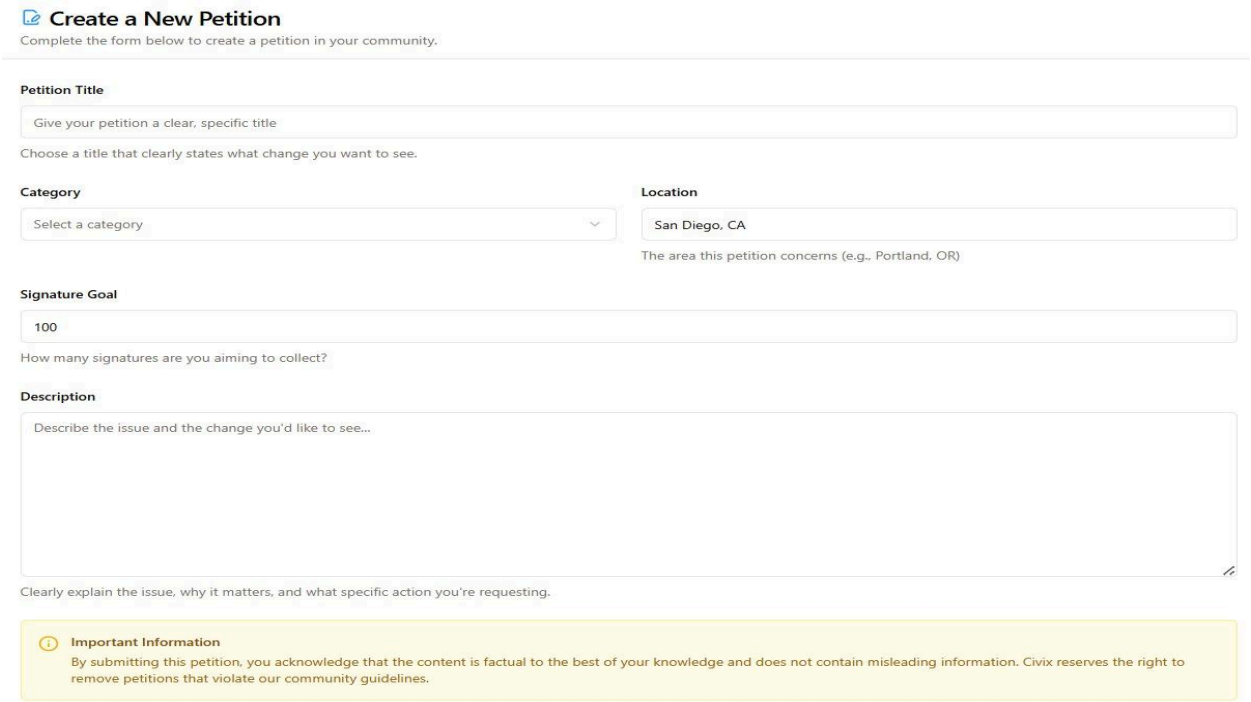




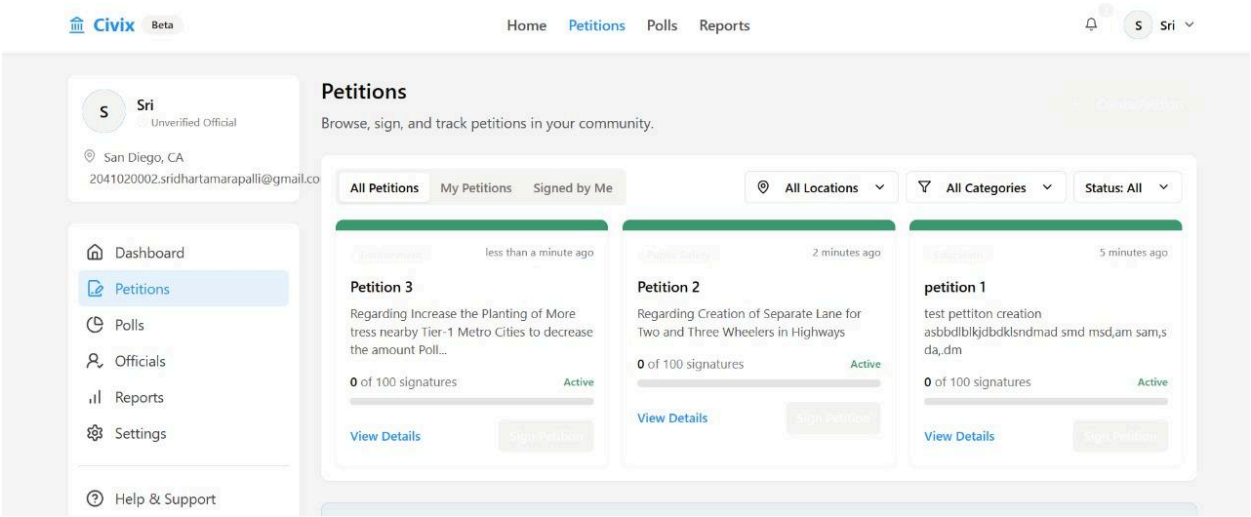
Dashboard screen -



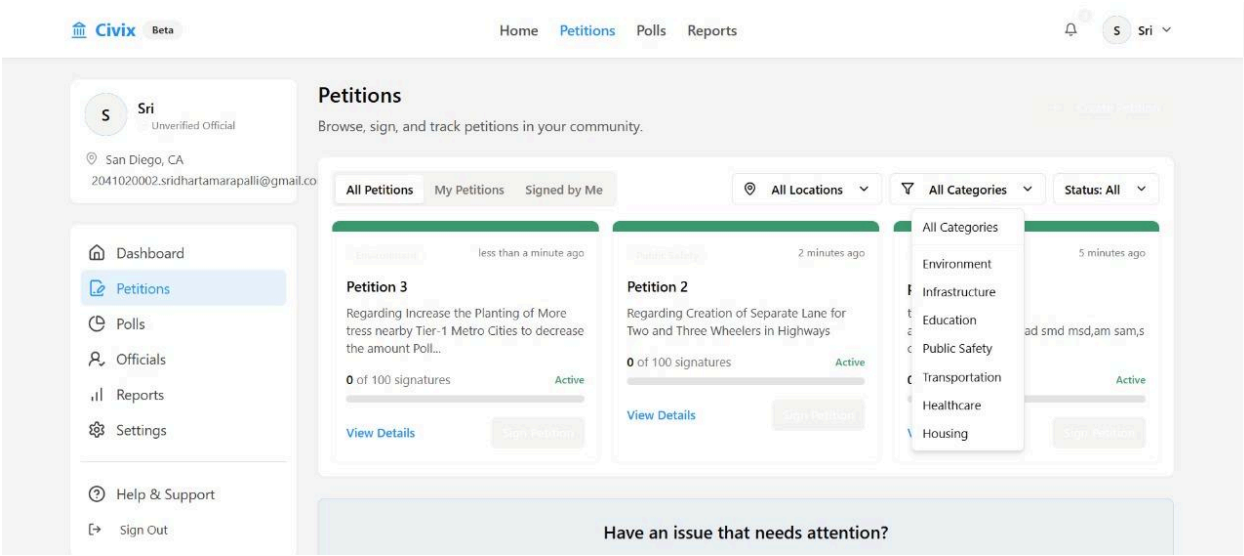
Petition creation -



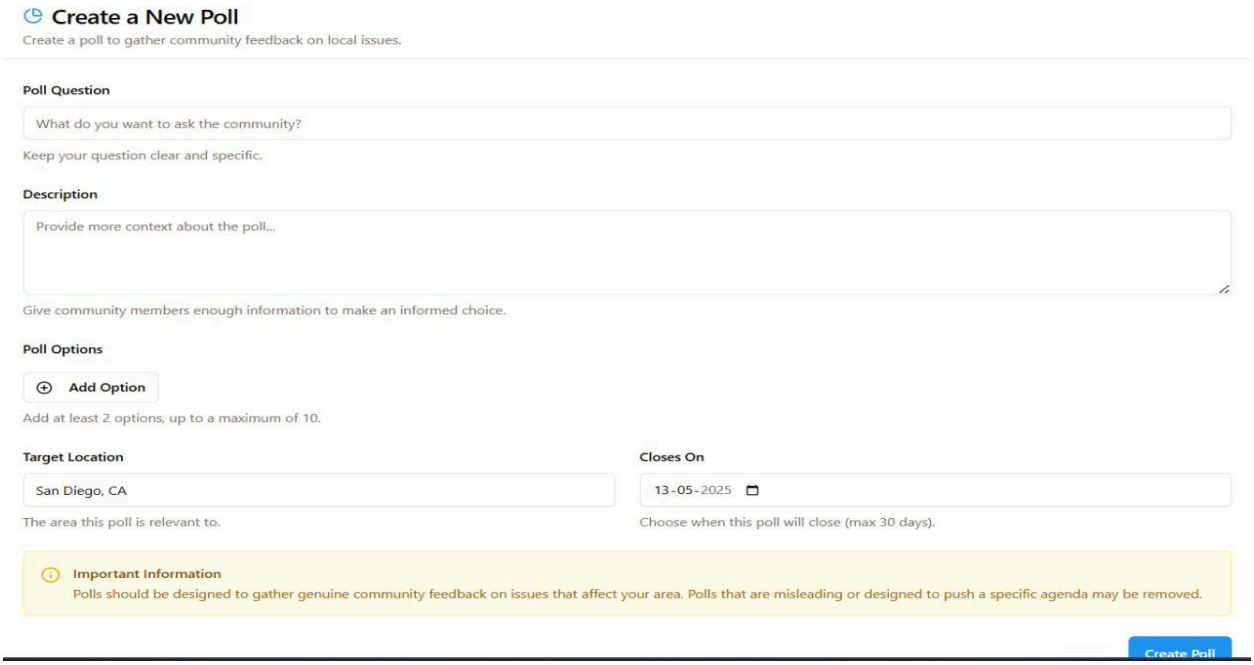
Petition list -



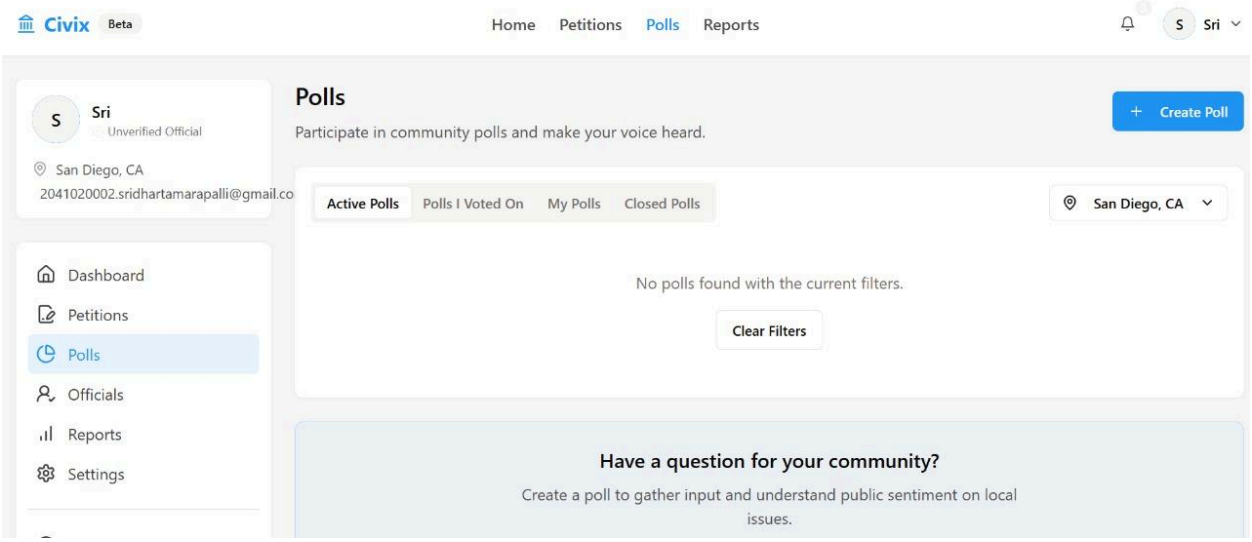
Petition Filter View-



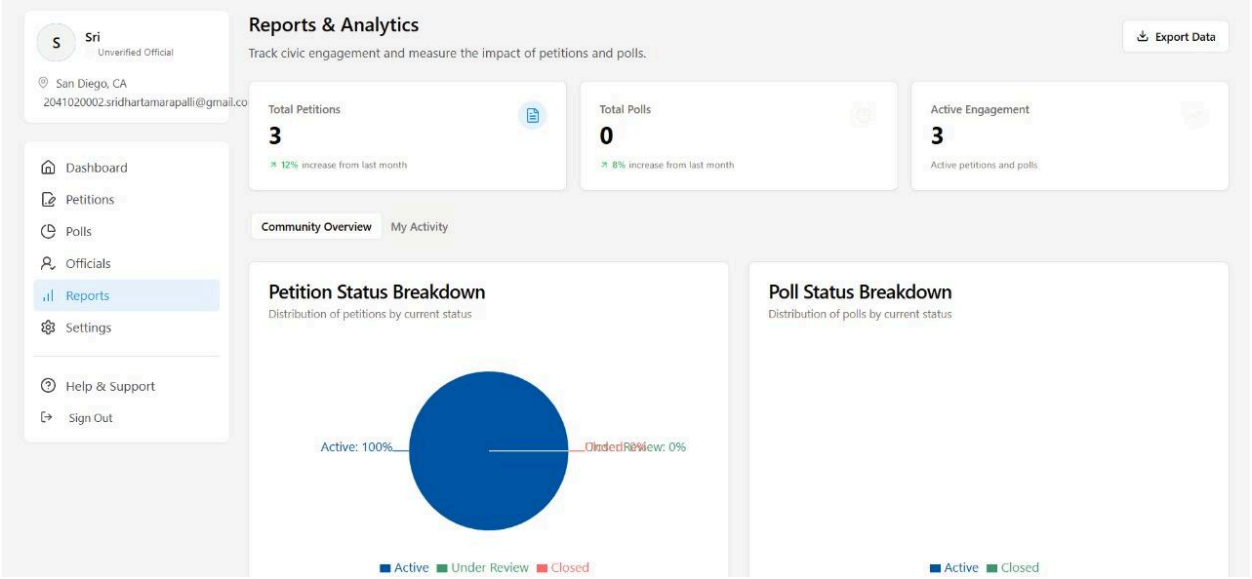
Poll Creation -



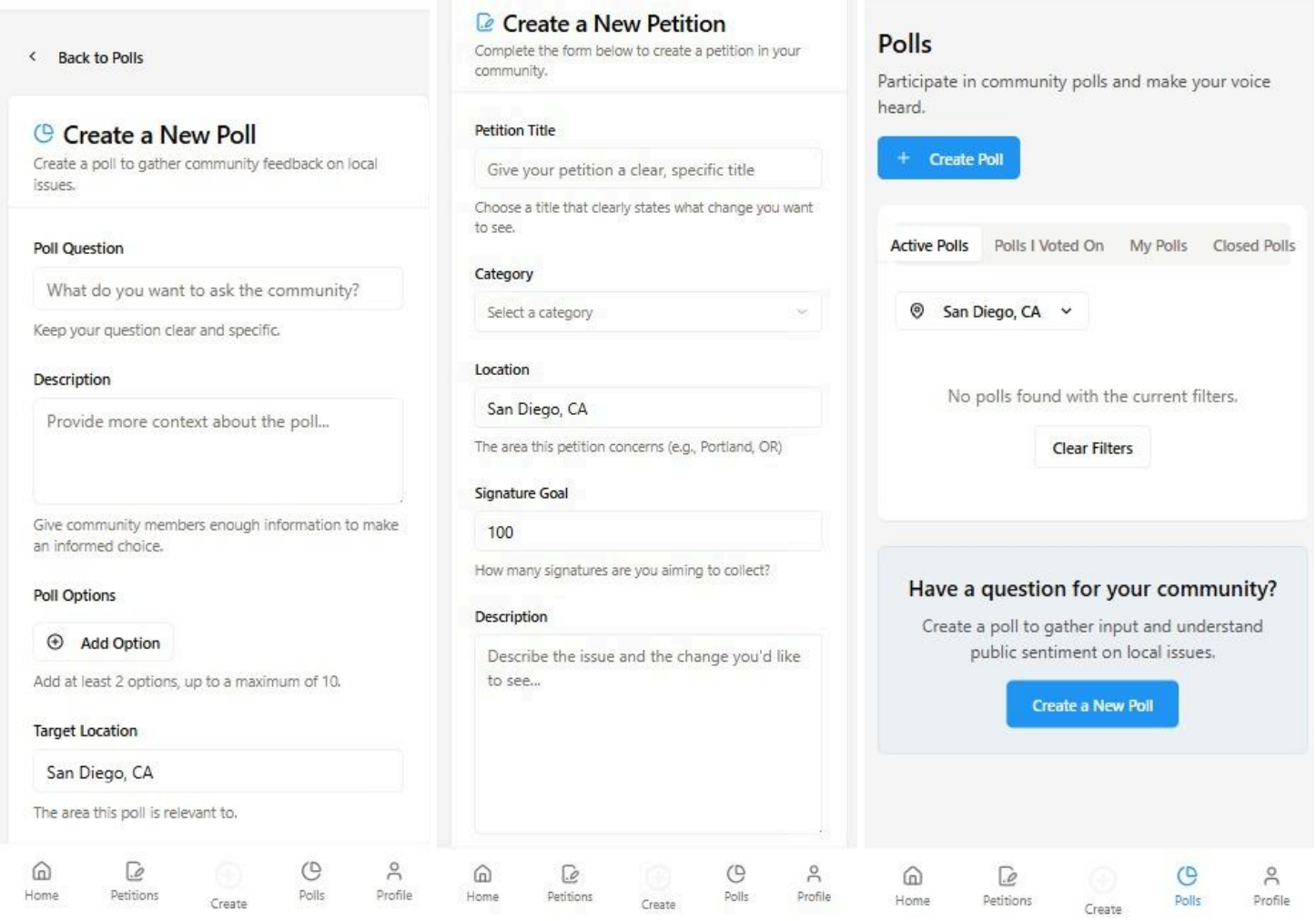
Poll list -



Reports dashboard with export button -



Responsive images -



My Petitions

2

petitions

Successful Petitions

0

or under review

Polls Created

0

polls

Active Petitions Near You

Showing for: San Diego, CA

All Categories | Environment | Infrastructure | Education | Public Safety | Transportation | Healthcare | Housing

16 minutes ago

Petition 3

Regarding Increase the Planting of More tress nearby Tier-1 Metro Cities to decrease the amount Poll...

Home | Petitions | Create | Polls | Profile

Welcome to Civix

Join our platform to make your voice heard in local governance.

Login | Register

Full Name

Jane Doe

Email

your@email.com

Password

••••••••

Location

Portland, OR

I am registering as:

Citizen | Public Official

Create Account

Already have an account? Sign in

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Full Name

Jane Doe

Email

your@email.com

Password

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Email

your@email.com

Password

••••••••

Sign In

Don't have an account? Register now

Reports & Analytics

Track civic engagement and measure the impact of petitions and polls.

Export Data

Total Petitions

3

↗ 12% increase from last month

Total Polls

0

↗ 8% increase from last month

Active Engagement

3

Active petitions and polls

Community Overview | My Activity

Petition Status Breakdown

Home | Petitions | Create | Polls | Profile